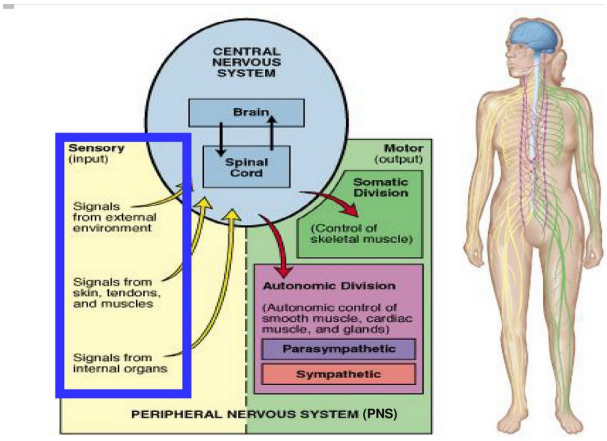
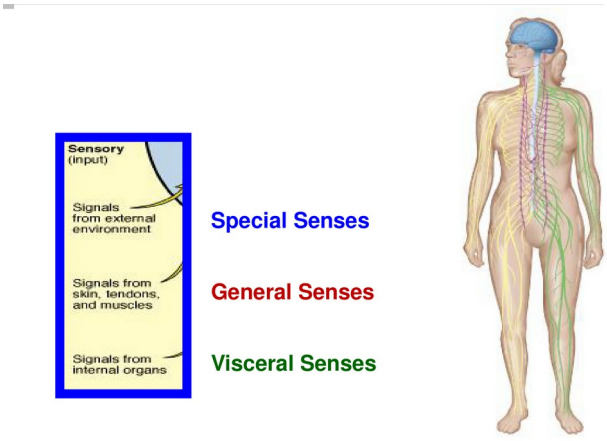
01 The sensory system: what enters the CNS

Sensory receptors detect the environment → neural pathways carry information → the central nervous system (CNS) interprets it.



Sensory input and motor output



Three sources of sensory information

Group	Receptor distribution	Examples
General / somatic senses	Scattered throughout the body wall, including skin, muscles and tendons.	Touch, pressure, temperature, pain and proprioception.
Visceral senses Splanchnesthesia	Internal organs within the thoracic and abdominal cavities.	Fullness of the bladder or stomach after a meal; visceral pain.
Special senses	Receptors concentrated in specialised organs or regions.	Vision, audition, vestibular sensation (balance), gustation and olfaction.

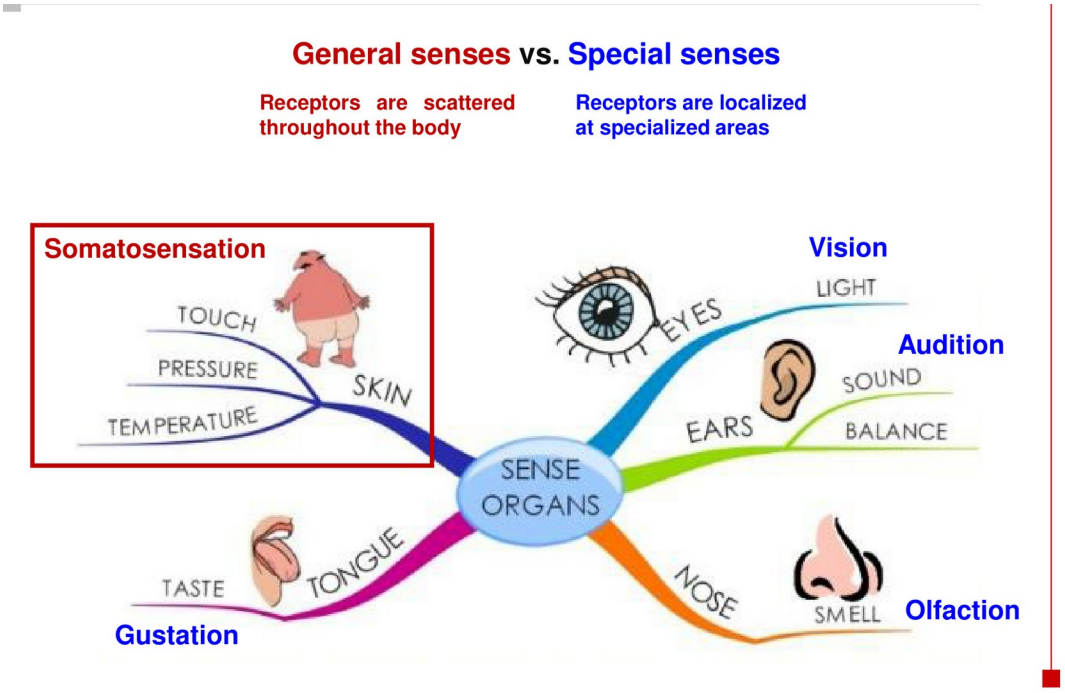
- The **peripheral nervous system (PNS)** carries sensory information toward the brain and spinal cord.
- Somatic motor output controls skeletal muscle. Autonomic motor output regulates smooth muscle, cardiac muscle and glands.
- Somatosensation describes sensation from the body; a visceral sensation concerns an internal organ.

02 Sensory receptors: stimulus specificity

A sensory receptor is a transducer: environmental energy
→ a change in electrical activity → a signal that the CNS can interpret.

Receptor class	Best stimulus	Examples
Mechanoreceptor	Compression, bending or stretching	Touch, pressure, proprioception, hearing and balance.
Thermoreceptor	Cooling or warming	Cold and warm sensations; temperature regulation.
Photoreceptor	Light	Rods and cones in the retina.
Chemoreceptor	Chemical substances	Taste, smell and blood O_2 / CO_2 sensing.
Nociceptor	Potentially damaging mechanical, chemical or thermal stimuli	Pain-related sensory input; free nerve endings.

- **Specificity:** each receptor type responds best to particular stimuli.
- Stimulus type and receptor location are different classifications: mechanoreceptors contribute to both general sensation and special senses.

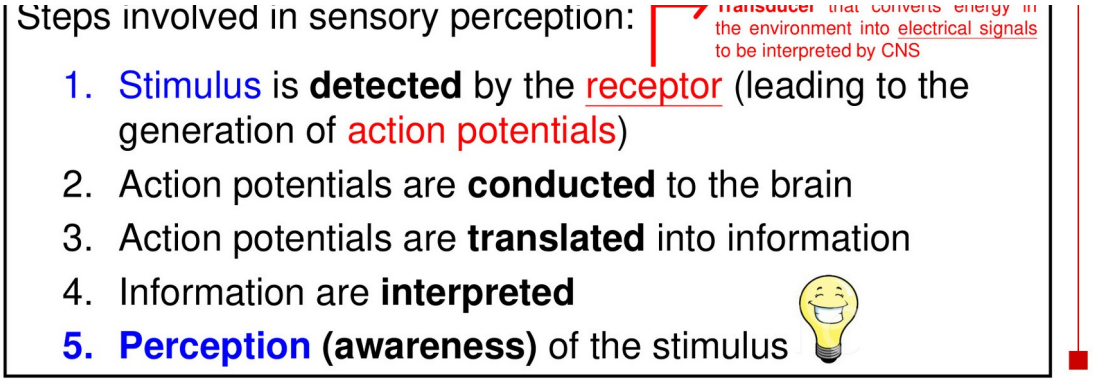


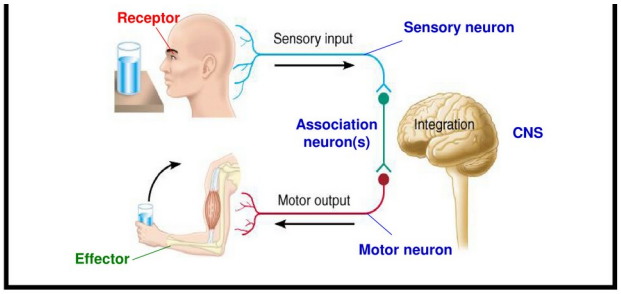
- General sensory receptors are distributed widely; special sensory receptors are localised in specialised areas.

03 Sensation vs perception

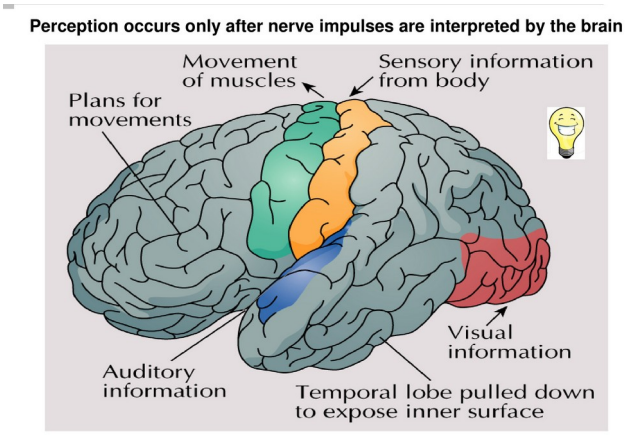
Sensation	Perception
Detection of a stimulus and transmission of the signal to the brain.	**Interpretation of sensory input and recognition of what is sensed.**
Begins at a receptor.	Depends on brain processing and is affected by experience.

Stimulus detected → electrical receptor response → sensory action potentials
→ impulses conducted to the brain → signals translated into information
→ information interpreted → **awareness of the stimulus**





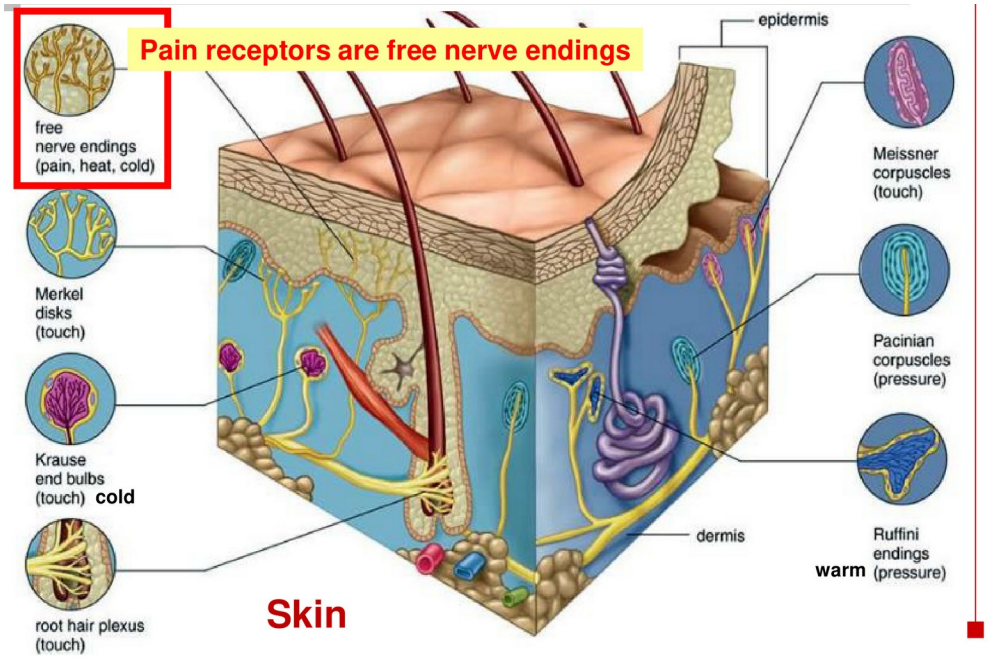
Input → integration → output



Different brain regions process different information.

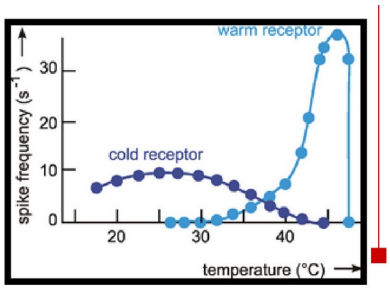
- **Perception occurs only after the brain interprets the sensory impulses.**
- Receptor cells need not generate action potentials themselves: rods and cones produce graded signals, while retinal ganglion cells carry the output as action potentials.
- Sensory input can also initiate motor output through reflex circuits.

04 Skin receptors & temperature sensing



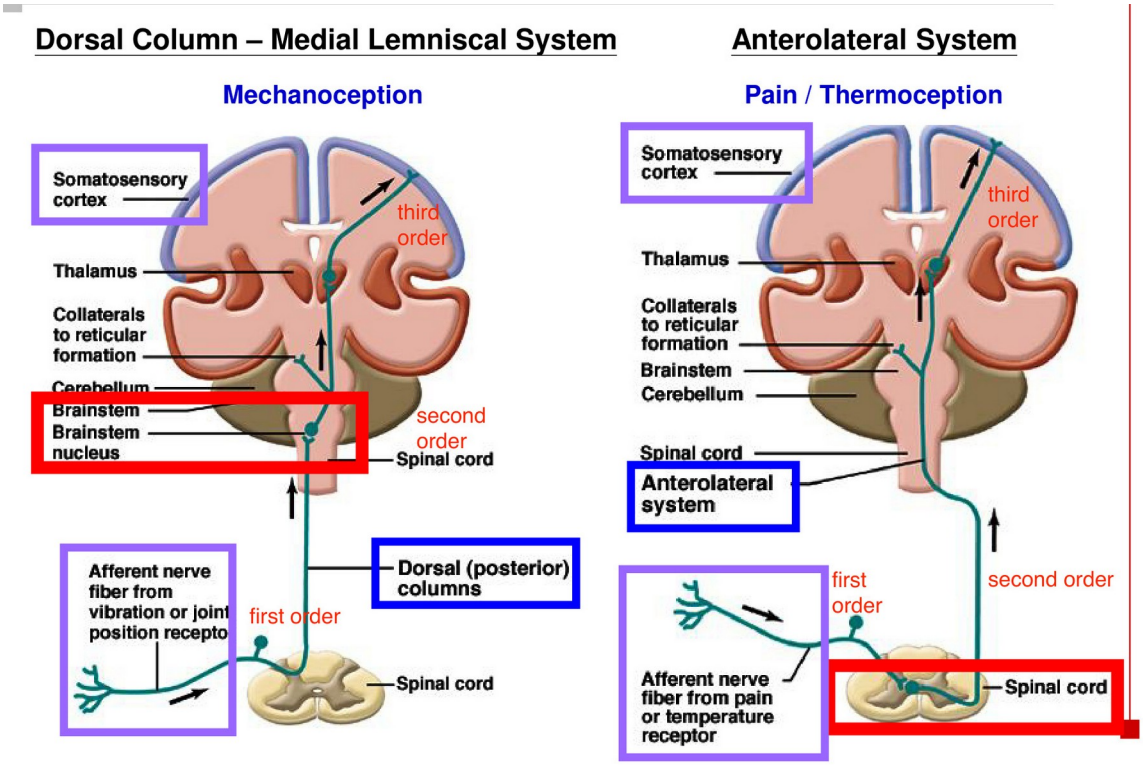
Cutaneous structure	Main association
Merkel discs / Meissner corpuscles / root hair plexus	Touch; hair movement is detected around the hair root.
Pacinian corpuscles	Pressure; deeper mechanical deformation and vibration.
Ruffini endings / Krause end bulbs	Named cutaneous endings in the receptor diagram. Ruffini endings detect skin stretch; Krause end bulbs are traditionally associated with touch / cold.
Free nerve endings	Pain, heat and cold. Nociceptors can respond to noxious mechanical, thermal or chemical stimuli.

Thermoreceptors	Location / role
Central	Hypothalamus → thermoregulation; operates without conscious temperature perception.
Cutaneous	Skin → conscious temperature sensation AND thermoregulation.
Cold / warm ranges	**Cold: 10–35°C. Warm: 30–45°C.** The response ranges overlap.



Temperature changes alter receptor firing. Modern temperature sensing centres on thermosensitive free endings; the traditional “Ruffini = warm / Krause = cold” labels are not exclusive functional identities.

05 Somatosensory pathways: where they cross



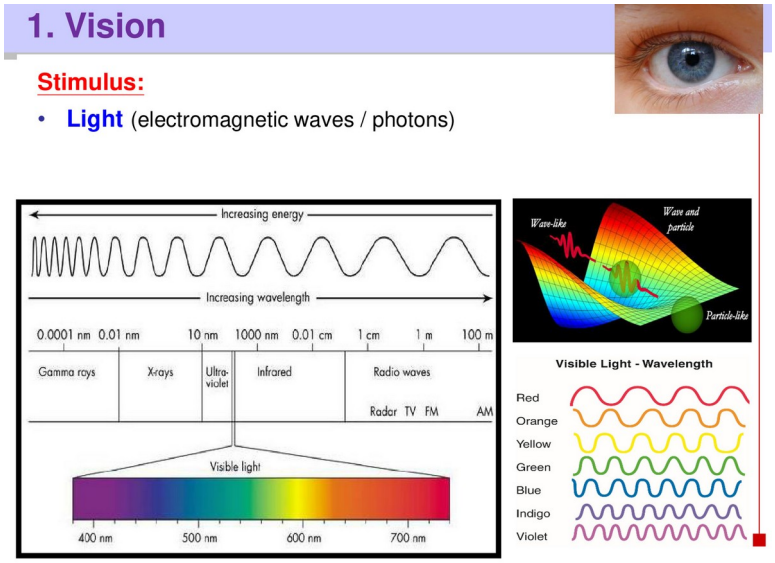
Feature	Dorsal column–medial lemniscal	Anterolateral / spinothalamic
Modalities	**Mechanoception:** discriminative touch, vibration and joint position.	**Pain, thermoception and crude touch.**
First-order neuron	Ascends on the same side in the dorsal (posterior) columns → brainstem nucleus.	Enters the spinal cord → synapses with a second-order neuron there.
Second-order neuron	**Crosses in the brainstem / medulla**, then ascends toward the thalamus.	**Crosses in the spinal cord**, then ascends in the opposite anterolateral pathway.
Third-order neuron	Thalamus → somatosensory cortex.	Thalamus → somatosensory cortex.

Dorsal column: ascend first, then cross in the brainstem.
Anterolateral: cross in the spinal cord, then ascend.

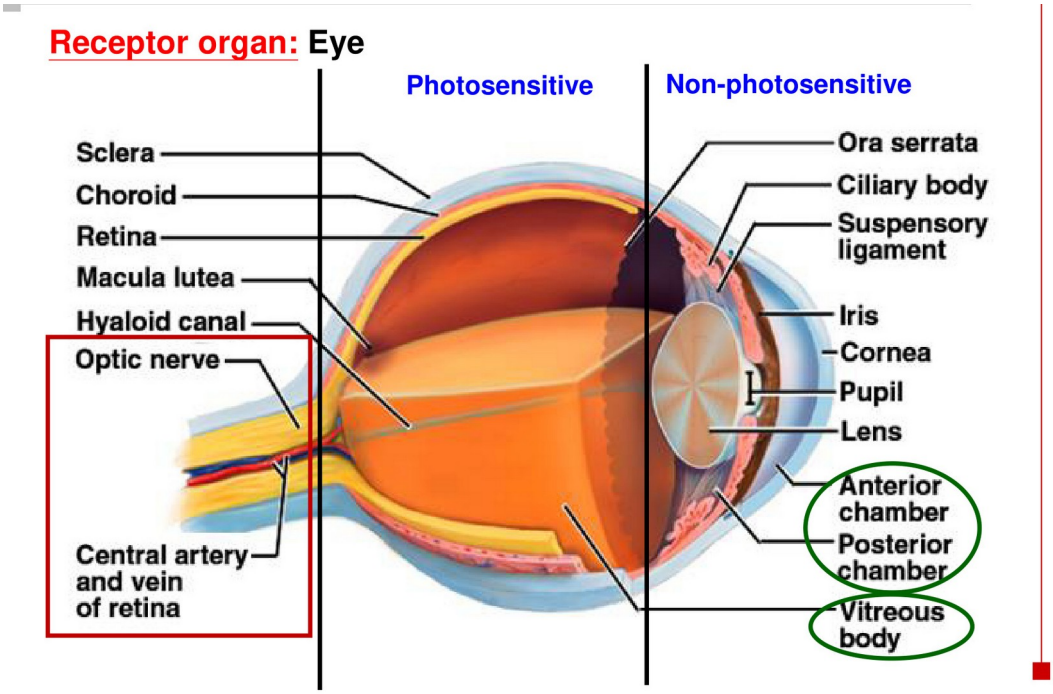
- Both pathways relay through the thalamus before reaching the somatosensory cortex. Collaterals also reach brainstem reticular circuits.

06 Vision: light & the eye

Stimulus: light (electromagnetic waves / photons).
Receptor organ: eye. Receptor cells: rods and cones. Receptor protein: opsin.



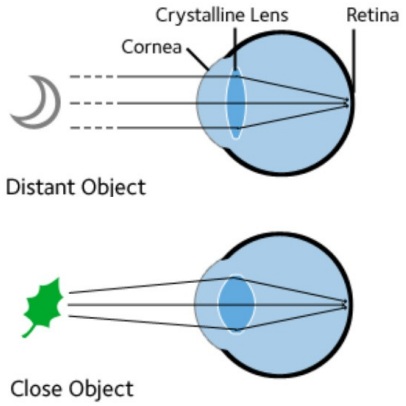
- Wavelength distinguishes different parts of the spectrum; visible light occupies approximately 400–700 nm. Shorter wavelength corresponds to higher photon energy.

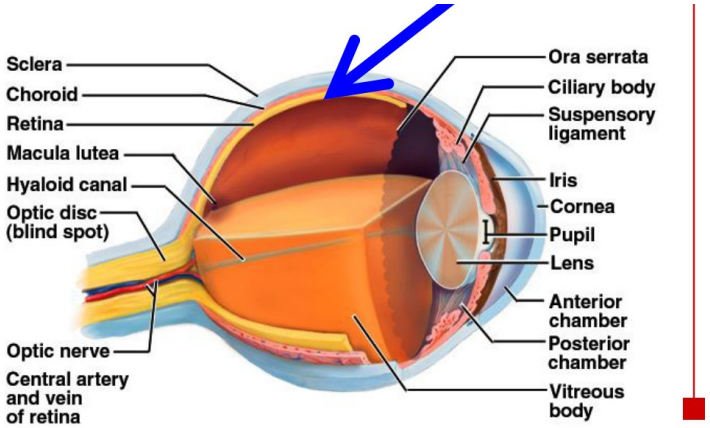


- The **retina is the innermost sensory layer**. Its posterior photosensitive region contains rods and cones; the anterior continuation is non-photosensitive.
- The ora serrata marks the anterior boundary of the photosensitive retina.

07 Eye structures: form supports function

Structure	Feature → function
Sclera	Tough outer coat → protects the eyeball and maintains its shape.
Choroid	Blood vessels → nutrient supply. Pigment → limits internal light reflection.
Cornea	Transparent anterior part of the outer layer → allows light to enter and contributes to refraction.
Pupil	Circular opening in the iris → admits light.
Iris	Coloured structure containing muscles → regulates pupil size.
Lens	Elastic and biconvex → focuses an image on the retina.
Ciliary body	Contains ciliary muscle → adjusts lens focusing through the suspensory ligaments.
Retina / optic nerve	Retina contains the sensory cells; ganglion-cell axons leave as the optic nerve.

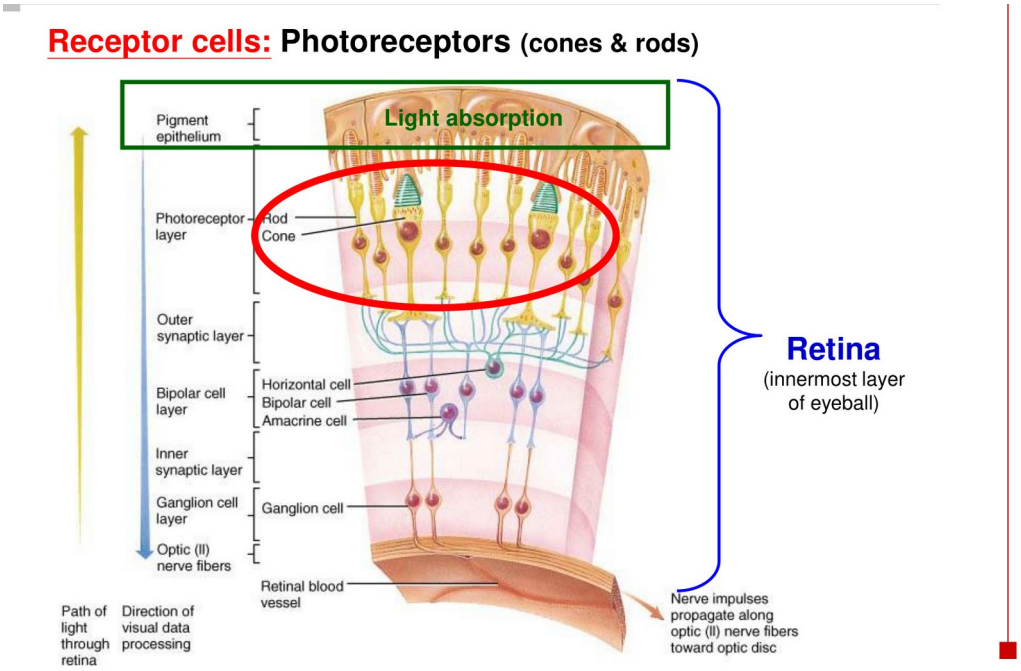




Light → **cornea → aqueous-filled anterior chamber → pupil → lens → vitreous body → retina**

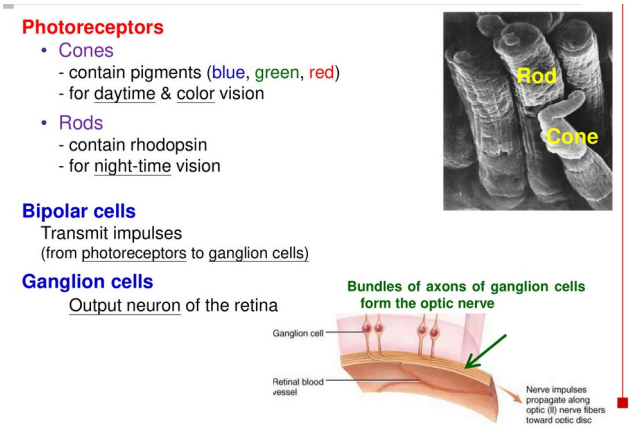
- The posterior chamber lies behind the iris and in front of the lens; the large vitreous compartment is behind the lens.
- Accommodation changes lens shape: a more rounded lens helps focus near objects; a flatter lens helps focus distant objects.
- Other landmarks: macula lutea, optic disc, central retinal artery and vein, and hyaloid canal within the vitreous body.

08 Retina: light direction vs signal direction



Light: passes through inner retinal layers → reaches rods and cones.
Neural signal: photoreceptors → bipolar cells → ganglion cells
→ **ganglion-cell axons form the optic nerve**.

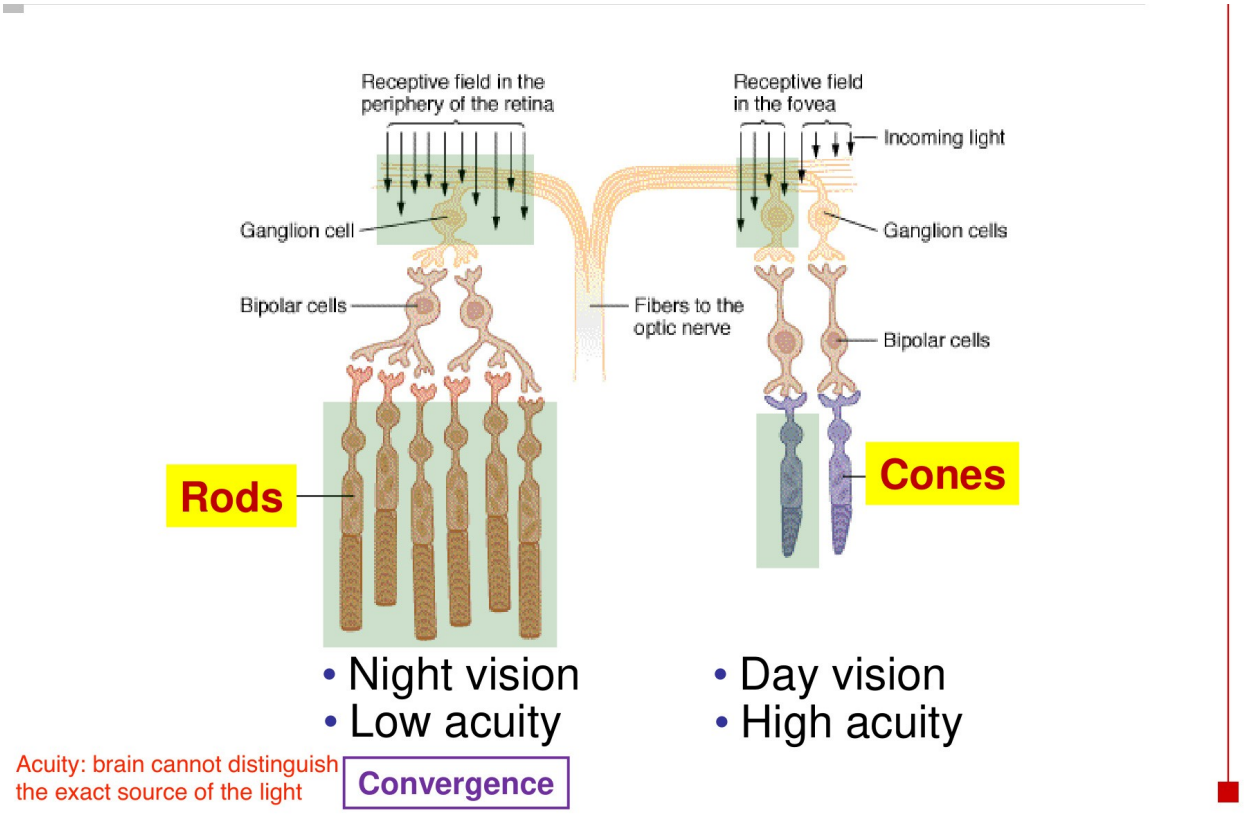
Cell / layer	Role
Rods and cones	Photoreceptors; absorb light near the pigment epithelium.
Bipolar cells	Relay changes in photoreceptor output to ganglion cells.
Ganglion cells	**Output neurons of the retina**; their axons conduct toward the brain.
Horizontal / amacrine cells	Lateral retinal connections that modify the signal between neighbouring pathways.



- The photoreceptor outer segment performs light transduction; the optic nerve carries processed retinal output.

09 Rods, cones & convergence

Feature	Rods	Cones
Vision	**Night / dim-light vision**	**Daylight and colour vision**
Photopigment	Rhodopsin	Cone pigments with blue-, green- and red-sensitive classes
Sensitivity	High sensitivity to weak light	Require brighter light for effective function
Convergence	Many rods feed into shared downstream pathways	Lower convergence, especially in the fovea
Spatial acuity	**Lower acuity**	**Higher acuity**

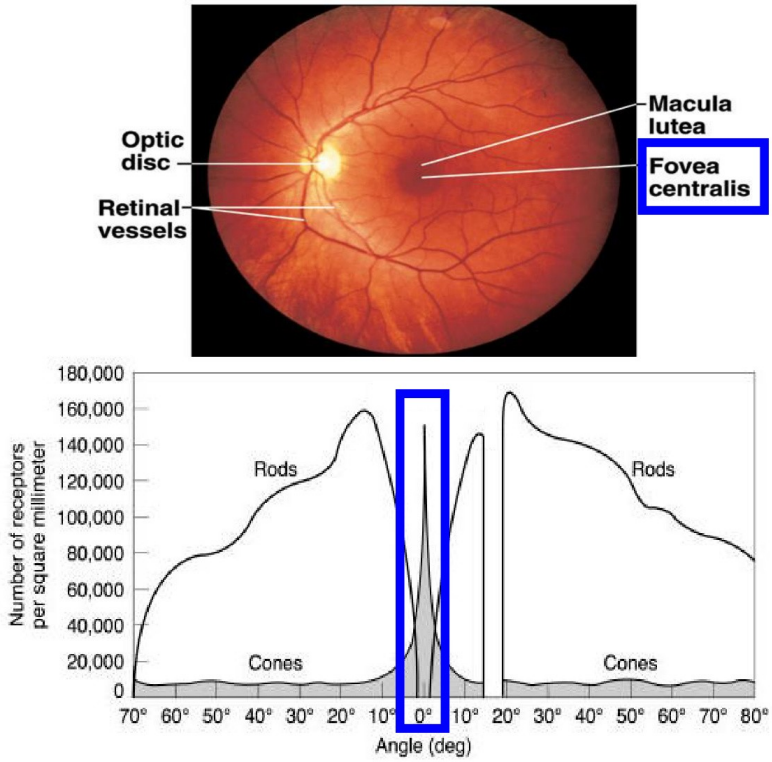


More convergence → signals from several receptors are pooled
→ better detection of weak input, but less certainty about its exact location.

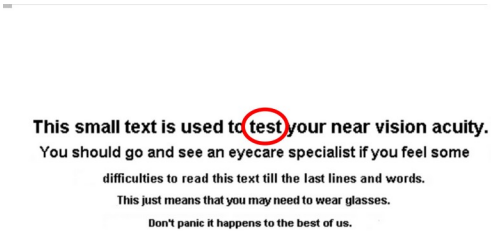
- If several rods share one output pathway, the brain cannot identify which particular rod detected the light.
- Fewer converging receptors preserve spatial detail: neighbouring points can remain distinguishable.

10 Fovea, acuity & the blind spot

Fovea: most acute image formation



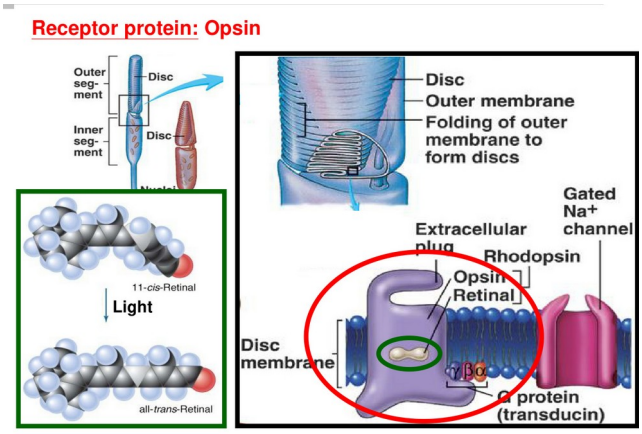
Landmark	What matters
Fovea centralis	Within the macula lutea; dense cones and low convergence → **most acute image formation**.
Peripheral retina	Rods are abundant outside the fovea; rod density peaks away from its centre.
Optic disc	Ganglion-cell axons leave the eye here. No rods or cones → physiological blind spot.



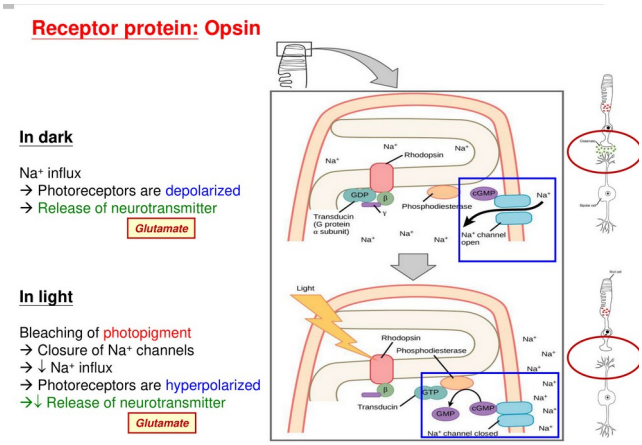
- Visual acuity means resolving fine spatial detail, such as small letters or neighbouring points.
- Image focus, illumination and viewing distance affect performance on a near-vision task.

11 Phototransduction: dark vs light

Dark: Na^+ influx → photoreceptor depolarisation → ongoing glutamate release.
Light: photopigment activation / bleaching → Na^+ channels close
→ Na^+ influx ↓ → hyperpolarisation → glutamate release ↓.



Opsin + retinal form the photopigment.



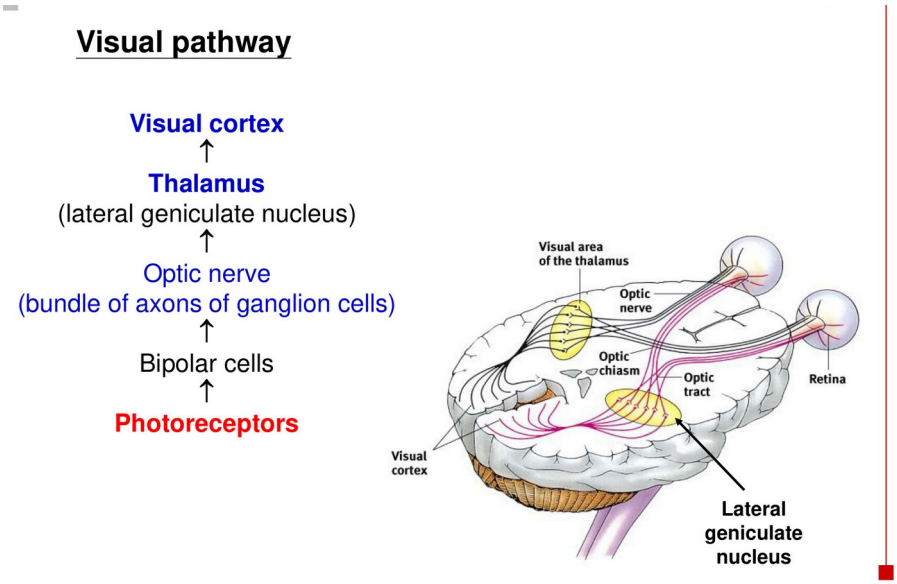
Light reduces inward current and transmitter release.

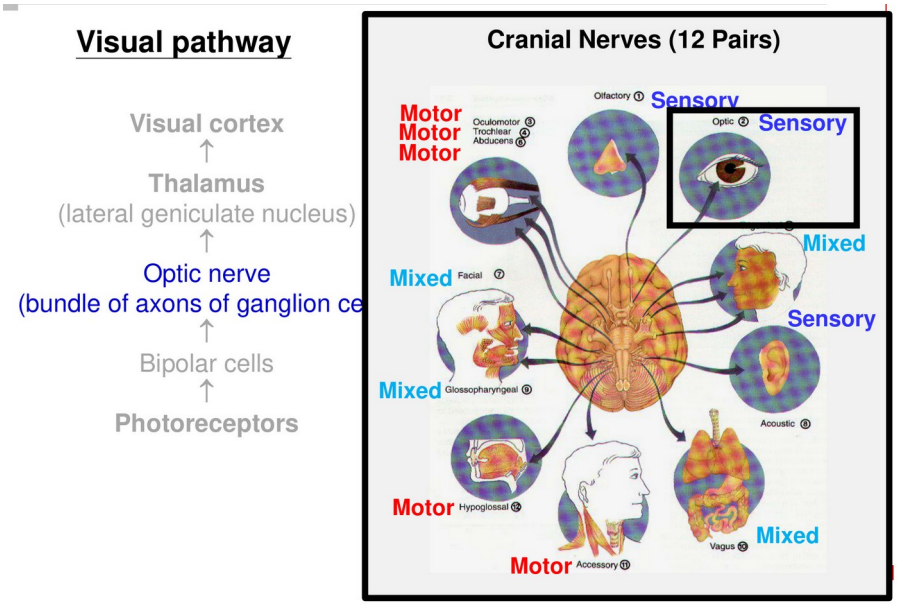
Key component	Role in the sequence
Outer-segment discs	Provide membrane area containing photopigment.
Opsin + retinal	Rhodopsin consists of rod opsin plus retinal. Light converts **11-cis retinal to all-trans retinal**.
Transducin	G protein activated downstream of the light-sensitive pigment.
Channel closure	The signalling cascade activates phosphodiesterase → cGMP falls → cyclic-nucleotide-gated cation channels close.
Glutamate	Released at the photoreceptor synapse; release is lower in light than in darkness.

- **Photoreceptors hyperpolarise in response to light.** This is the opposite of a simple "stimulus opens inward channels → depolarisation" model.
- Photoreceptor voltage and transmitter release change gradually; retinal ganglion cells provide the action-potential output.

12 Visual pathway: retina to cortex

Rods / cones → bipolar cells → ganglion cells
→ optic nerve (CN II) → optic chiasm → optic tract
→ lateral geniculate nucleus (LGN) of the thalamus
→ optic radiations → visual cortex in the occipital lobe





- **The optic nerve is a bundle of ganglion-cell axons**, not photoreceptor or bipolar-cell axons.
- CN II is sensory. The thalamic visual relay is the **LGN**.
- The optic nerve is before the chiasm; the optic tract is after it. Optic radiations connect the LGN to visual cortex.

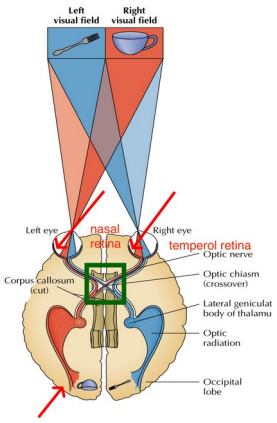
13 Visual fields & retinotopic organisation

Two optic nerves meet at the **optic chiasm**

- Only fibers from nasal half of retina cross over

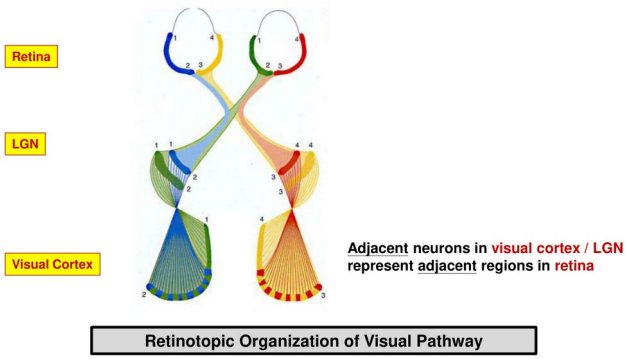
Signals from left visual field are projected to right visual cortex

Signals from right visual field are projected to left visual cortex



Only nasal retinal fibres cross at the chiasm.

Spatial relationships among ganglion cells in retina are maintained in their central targets as orderly representations ("maps") of visual space



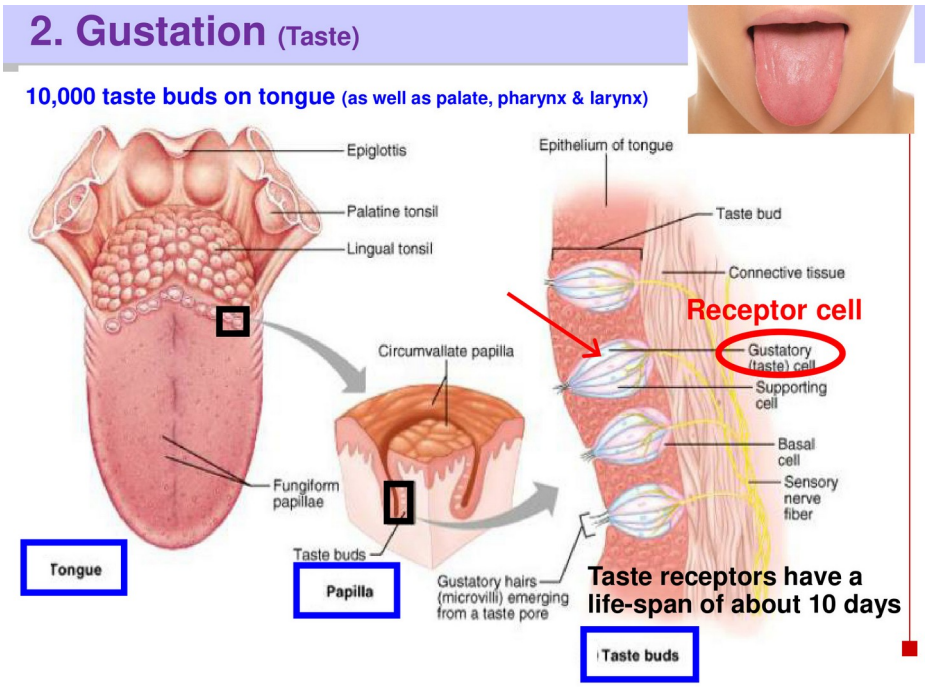
Spatial relationships are preserved in central maps.

Rule	Consequence
Two optic nerves meet at the optic chiasm.	Fibres from the **nasal half of each retina cross**; temporal retinal fibres remain on the same side.
Left visual field → right visual cortex.	Left-eye nasal retina and right-eye temporal retina supply the right-sided pathway.
Right visual field → left visual cortex.	Right-eye nasal retina and left-eye temporal retina supply the left-sided pathway.
Retinotopy	Spatial relationships among retinal ganglion cells are retained as orderly maps in the LGN and visual cortex.

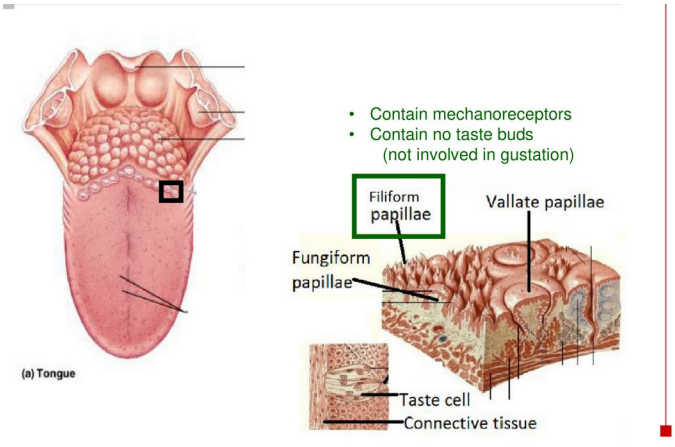
- A visual field is a region of external space; it is not the same as "the left eye" or "the right eye". Both eyes contribute to each side's visual-field representation.
- **Adjacent neurons in the LGN / visual cortex represent adjacent retinal regions.**

Optic chiasm = selective crossing.
Retinotopic organisation = preservation of spatial relationships.

14 Taste buds & papillae

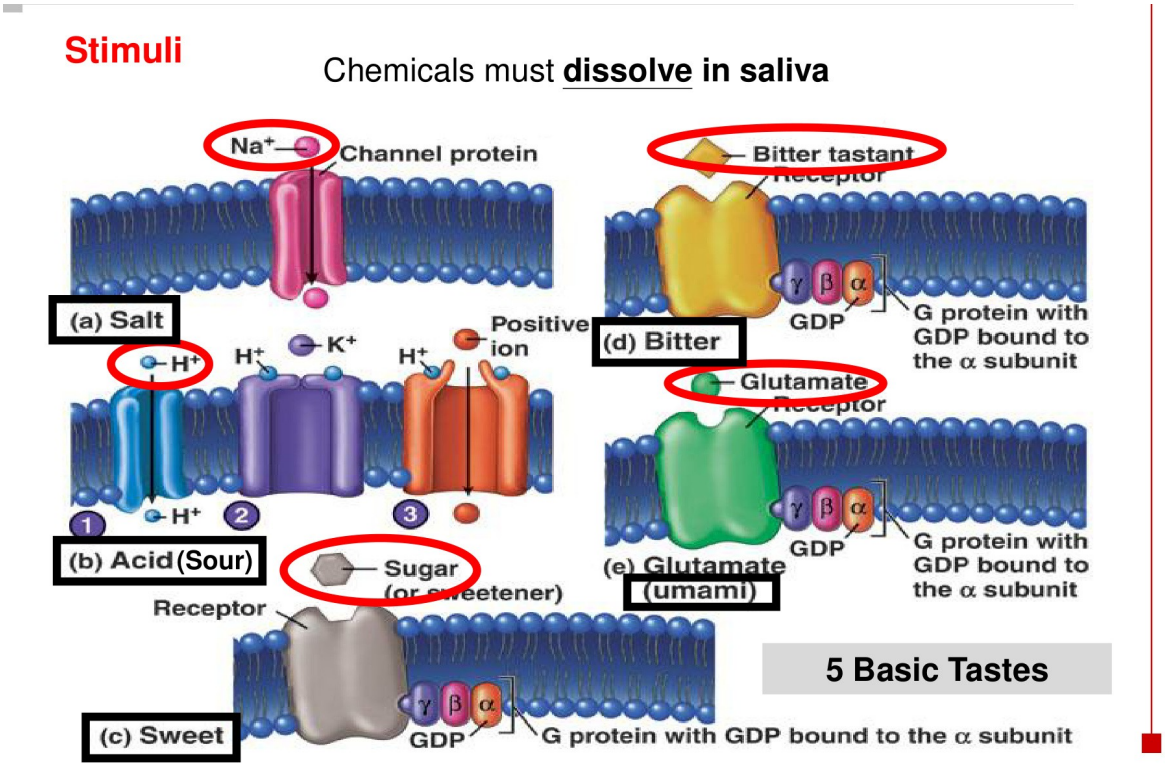


Structure / fact	Essential detail
Taste-bud distribution	Approximately **10,000 taste buds**; located mainly on the tongue, with additional buds in the palate, pharynx and larynx.
Papillae	Surface structures of the tongue. Fungiform and circumvallate (vallate) papillae contain taste buds.
Filiform papillae	Contain mechanoreceptors but **no taste buds** → mechanical / tactile role rather than gustation.
Taste-bud cells	Gustatory receptor cells, supporting cells and basal cells; sensory nerve fibres contact the receptor cells.
Taste pore / gustatory hairs	Microvilli project through the pore to contact dissolved chemicals.
Continual turnover	Taste receptor cells have a lifespan of approximately **10 days** and are continually replaced.



15 Five tastes: two entry mechanisms

Tastant → **dissolves in saliva** → reaches the receptor surface
→ interacts with an ion channel OR binds a G-protein-coupled receptor
→ receptor-cell electrical change → sensory signalling



Taste	Example stimulus	Initial interaction
Salty	Na^+	Ion-channel pathway; inward positive charge promotes depolarisation.
Sour	H^+ from acids	H^+ entry, cation-channel opening or K^+-channel inhibition.
Sweet	Sugar / sweetener	G-protein-coupled receptor (GPCR).
Bitter	Bitter tastant	GPCR.
Umami	Glutamate	GPCR; savoury taste.

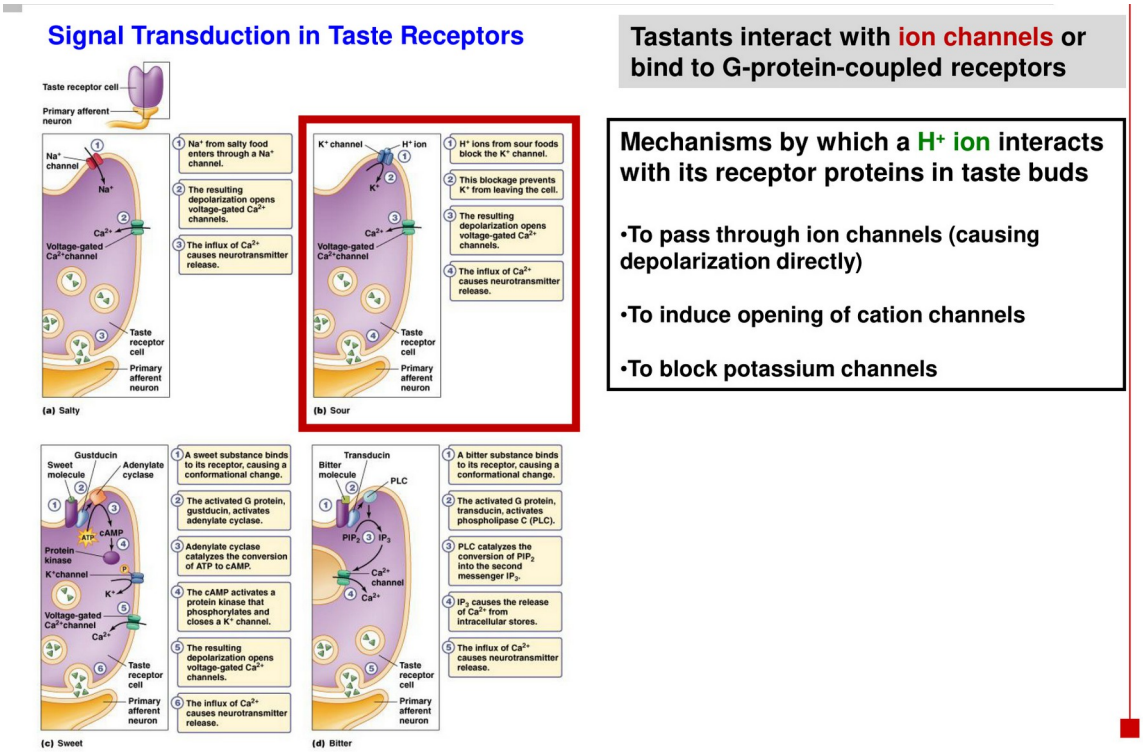
- **Taste quality depends on the receptor pathway activated**, not simply on whether a cell depolarises.
- Glutamate is an umami tastant in food; in the retina it serves as a neurotransmitter. The location and role differ.

16 Sour & salty: ion movements

Three ways H^+ can promote a sour response

H^+ action	Electrical consequence
1. Passes through an ion channel	Positive charge enters → direct depolarisation.
2. Opens a cation channel	Inward cation current increases → depolarisation.
3. Blocks a K^+ channel	Less K^+ leaves the cell → less positive charge is lost → depolarisation.

Sour stimulus → **taste receptor depolarises**
→ voltage-gated Ca^{2+} channels open → Ca^{2+} enters
→ neurotransmitter release → primary sensory afferent activated



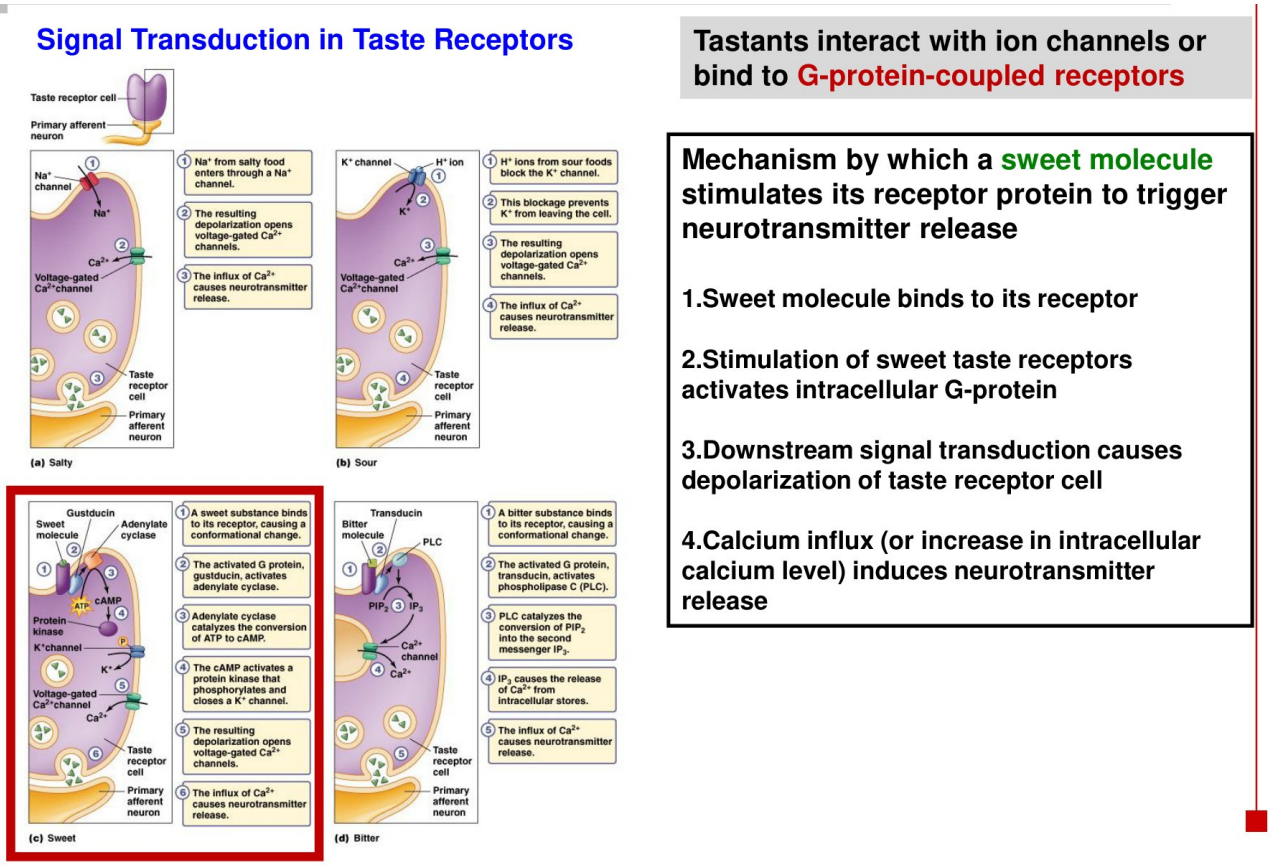
Salty pathway

Na^+ from salty food enters through a Na^+ channel → **depolarisation**
→ voltage-gated Ca^{2+} channels open → Ca^{2+} influx
→ neurotransmitter release → sensory afferent

- Blocking outward K^+ movement and increasing inward positive charge can both make the membrane less negative.

17 Sweet taste: receptor to transmitter

Sweet molecule binds its receptor
→ intracellular G protein activated
→ downstream signalling depolarises the taste receptor cell
→ Ca^{2+} influx and/or intracellular Ca^{2+} ↑
→ neurotransmitter release → sensory afferent activation



Tastants interact with ion channels or bind to G-protein-coupled receptors

Mechanism by which a sweet molecule stimulates its receptor protein to trigger neurotransmitter release

1.Sweet molecule binds to its receptor

2.Stimulation of sweet taste receptors activates intracellular G-protein

3.Downstream signal transduction causes depolarization of taste receptor cell

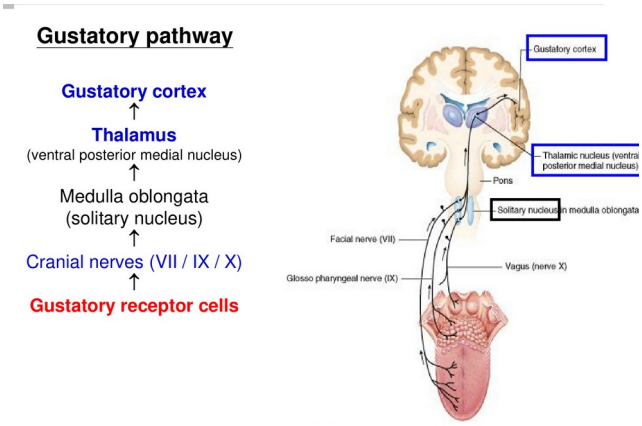
4.Calcium influx (or increase in intracellular calcium level) induces neurotransmitter release

Stage	Meaning
Recognition	The extracellular tastant binds a membrane receptor.
Transduction	The receptor activates an intracellular G protein; downstream signalling changes ion-channel activity.
Electrical response	The taste receptor cell depolarises.
Output	A rise in intracellular Ca^{2+} supports transmitter release to the sensory pathway.

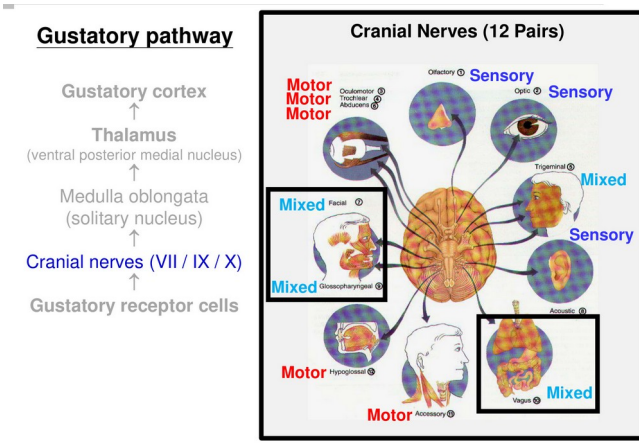
- **Sweet, bitter and umami use GPCR-based detection.** Salty and sour are introduced through ion-channel mechanisms.
- A G protein links receptor activation to intracellular signalling; the tastant itself does not need to enter the sensory nerve.

18 Gustatory pathway: tongue to insula

Gustatory receptor cells in taste buds
→ cranial nerves VII / IX / X
→ solitary nucleus in the medulla oblongata
→ ventral posterior medial nucleus (VPM) of the thalamus
→ gustatory cortex in the insula



Brainstem → thalamus → gustatory cortex



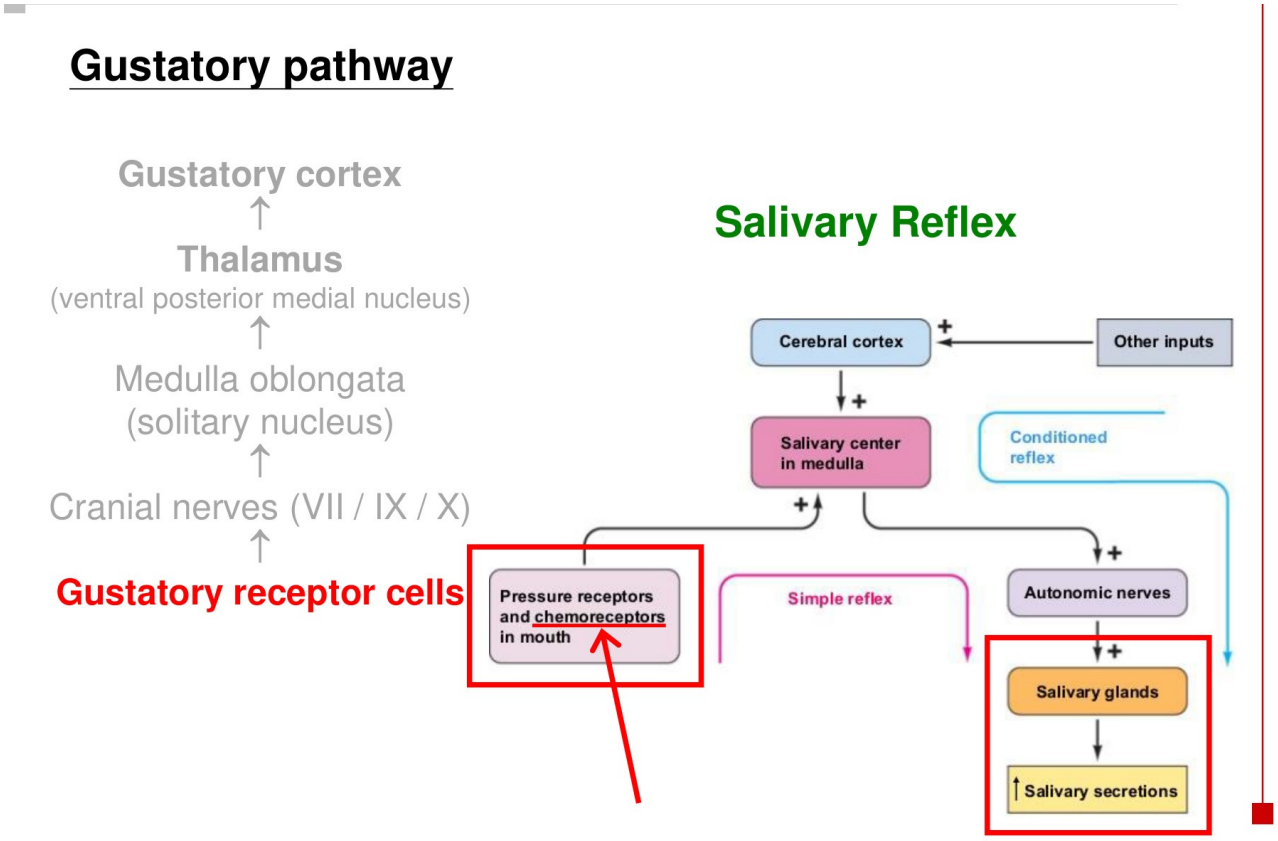
Taste information travels through VII, IX and X.

Cranial nerve	Name	Main taste territory
VII	Facial	Anterior part of the tongue.
IX	Glossopharyngeal	Posterior part of the tongue.
X	Vagus	Epiglottic / lower pharyngeal region.

- VII, IX and X carry sensory taste signals even though each nerve also has other functions and is classified as mixed.
- **VPM = taste relay; LGN = visual relay.** Both are thalamic nuclei, but they serve different sensory pathways.

19 Gustatory input & salivation

Taste chemicals / pressure in the mouth → oral receptors
→ sensory afferent input → brainstem salivary centre
→ autonomic nerves → salivary glands → salivary secretion ↑

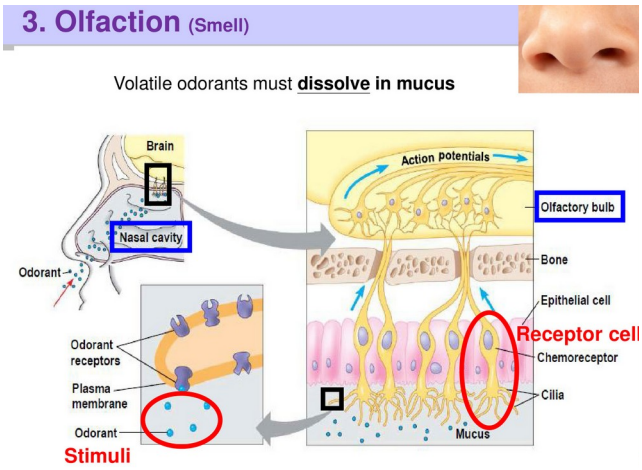


Route	Trigger → integration → effect
Simple salivary reflex	Oral chemoreceptors and pressure receptors → brainstem circuit → autonomic stimulation of salivary glands.
Conditioned salivary reflex	Other sensory / learned food-related inputs → cerebral cortex → salivary centre → glandular output.
Conscious taste perception	Receptor input → solitary nucleus → VPM thalamus → insular gustatory cortex.

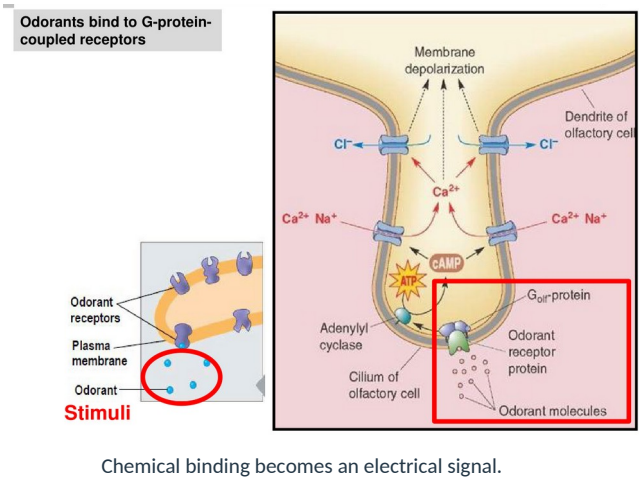
- **Gustatory input can produce both perception and a reflex response.** The salivary reflex does not require the full cortical recognition step first.
- Saliva helps dissolve tastants so that they can contact taste receptors.

20 Olfaction: receptor & transduction

Volatile odorant → **dissolves in nasal mucus**
→ binds an **olfactory GPCR** on receptor-cell cilia
→ G-protein signalling → cation channels open → Na^+ / Ca^{2+} entry
→ **depolarisation → action potentials**



Receptor neurons lie in olfactory epithelium.



Chemical binding becomes an electrical signal.

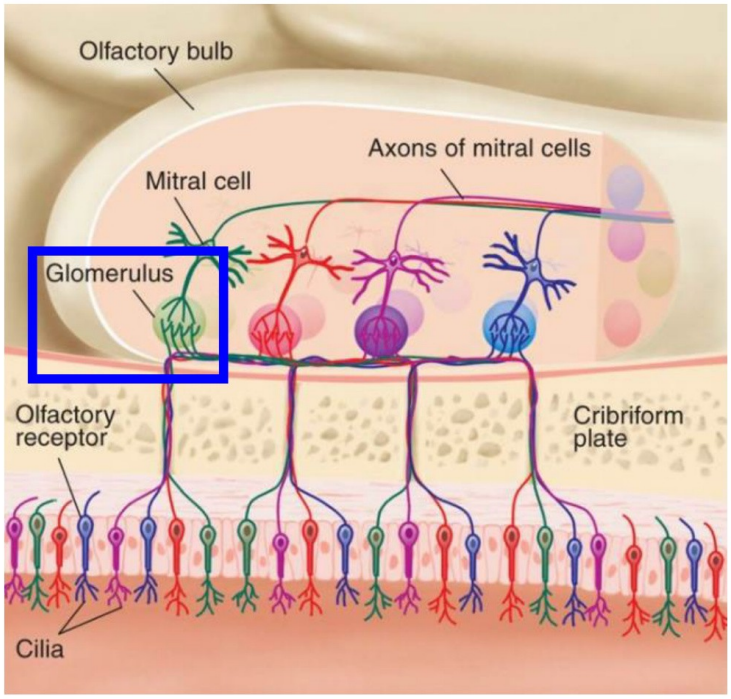
Structure / molecule	Role
Olfactory epithelium	Located inside the nasal cavity; contains olfactory chemoreceptor neurons.
Cilia and mucus	Receptor-cell processes extend into mucus, where dissolved odorants contact receptor proteins.
Odorant receptor / G protein	GPCR activation starts the intracellular signal leading to channel opening.
Na^+ and Ca^{2+} influx	Positive charge enters → depolarisation.
Ca^{2+}-activated Cl^- current	Cl^- leaves these receptor cells → loss of negative charge further supports depolarisation.

- **Taste requires dissolution in saliva; smell requires dissolution in mucus.** Both are chemical senses.

21 Olfactory bulb: glomeruli & the odour map

Olfactory receptor neurons → axons pass toward the olfactory bulb
→ **synaptic relay in glomeruli** → mitral-cell output
→ olfactory tract → central olfactory targets

Signals from olfactory receptors are relayed at glomeruli in olfactory bulb



Glomeruli form the "odor map"

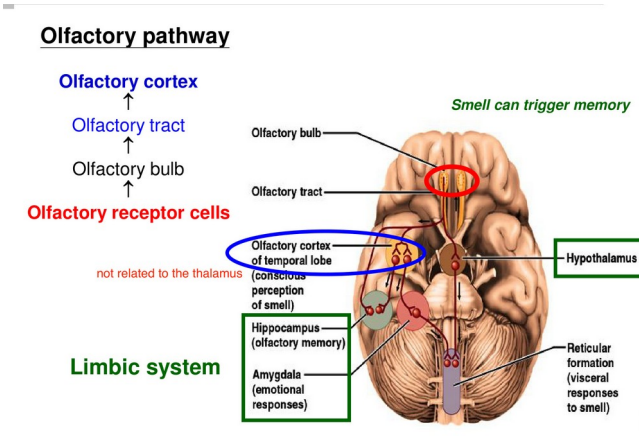
Each odor activates a specific pattern of glomeruli

Component	What it contributes
Receptor neurons	Detect dissolved odorants; carry the signal from the nasal epithelium.
Glomeruli	Organised synaptic relay sites in the olfactory bulb; receive converging receptor-neuron input.
Mitral cells	Receive input in the bulb and send output axons into the olfactory tract.
Odour map	Each odour activates a characteristic **pattern of glomeruli**.

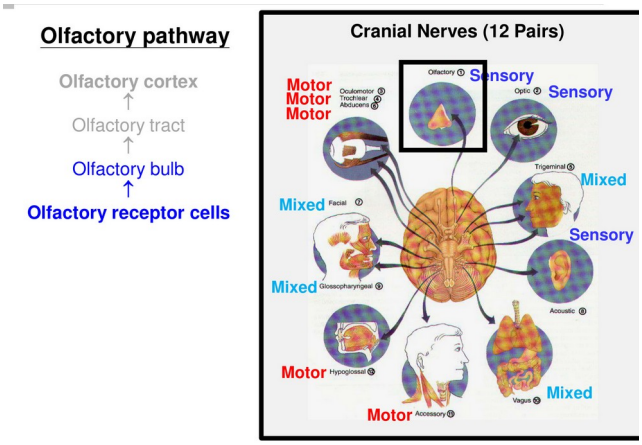
- Odour identity is represented by a pattern of activity, rather than requiring one isolated glomerulus for each whole odour.
- The glomerular relay is the first prominent central processing stage of the olfactory pathway.

22 Olfactory pathway, memory & emotion

Olfactory receptor cells → olfactory bulb → olfactory tract → primary olfactory cortex in the temporal lobe



Cortical and limbic connections



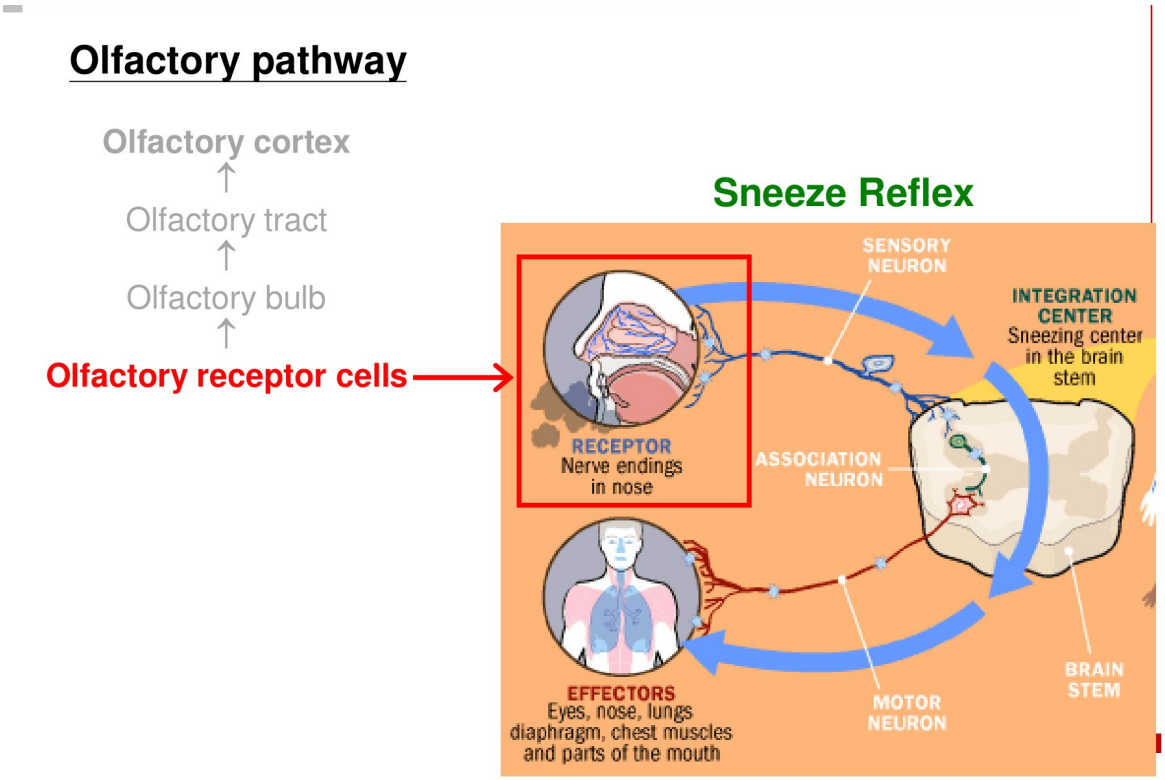
CN I is the sensory olfactory nerve.

Target / feature	Functional link
Olfactory cortex	Cortical processing of smell.
Limbic system	Links odour signals with memory and emotional responses.
Hippocampus	Olfactory memory associations.
Amygdala	Emotional responses to smell.
Hypothalamus / reticular formation	Visceral and autonomic responses associated with smell.
Thalamic distinction	There is **no obligatory thalamic relay before primary olfactory cortex**, unlike the visual and gustatory pathways.

- **Smell can trigger a memory** because olfactory processing communicates closely with limbic networks.
- CN I = olfactory, sensory. CN II = optic, sensory. Taste travels through VII / IX / X.

23 Nasal input & the sneeze reflex

Odorous irritants / other nasal irritants → **nasal sensory nerve endings**
→ sensory afferents → **brainstem sneeze centre**
→ coordinated motor output → respiratory and facial muscle activity
→ **sneeze: forceful expulsion helps clear the irritant**



- Smell-associated nasal stimulation can accompany or trigger a protective sneeze response.
- The usual irritant afferent limb is **mainly trigeminal (CN V)**; conscious smell is carried through **CN I**. Chemical irritation and olfactory perception are related experiences with distinct sensory pathways.

Sensory input	Conscious pathway	Reflex / related response
Taste	Brainstem → VPM → insular gustatory cortex	Salivary secretion.
Smell / nasal chemicals	Olfactory bulb → tract → olfactory cortex	Memory / emotion; nasal irritation can trigger sneezing.

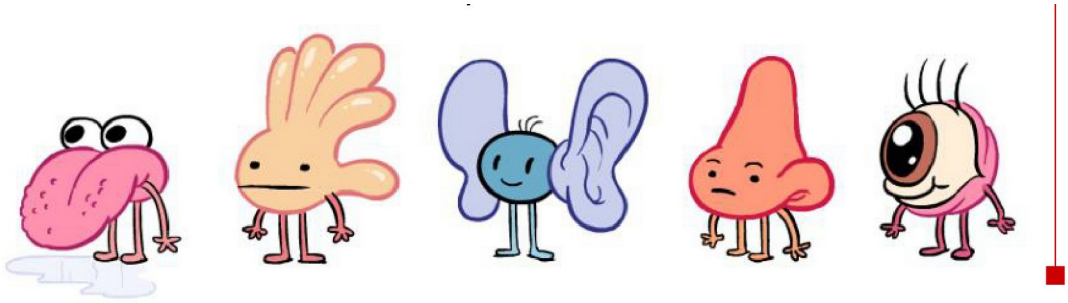
- The sneeze circuit coordinates association neurons and motor outputs to respiratory muscles, including the diaphragm and chest muscles, with facial / upper-airway actions.

24 Integrated revision: follow each signal

Modality	Receptor / entry signal	Main central route
Mechanoception	Mechanical deformation → mechanoreceptor response	Dorsal column → brainstem crossing → thalamus → somatosensory cortex.
Pain / temperature	Nociceptor / thermoreceptor input	Spinal synapse and crossing → anterolateral pathway → thalamus → somatosensory cortex.
Vision	Light → rods / cones **hyperpolarise → glutamate ↓**	Bipolar → ganglion → II → chiasm / tract → **LGN** → occipital cortex.
Taste	Dissolved tastant → channels / GPCRs → receptor-cell depolarisation and signalling	**VII / IX / X** → solitary nucleus → **VPM** → **insula**.
Smell	Odorant in mucus → olfactory GPCR → receptor-neuron depolarisation	**I → bulb glomeruli → tract → temporal olfactory cortex**; limbic connections.

Keep these distinctions clear

- **Sensation:** detect and transmit. **Perception:** interpret and recognise.
- **Rods:** dim light, convergence, lower acuity. **Cones:** daylight / colour, higher acuity.
- **Fovea:** highest acuity. **Optic disc:** no photoreceptors → blind spot.
- **Light reduces photoreceptor glutamate release.** It does not depolarise rods and cones.
- **Only nasal retinal fibres cross.** Each cerebral hemisphere represents the opposite visual field.
- **Salty / sour:** ion-channel mechanisms. **Sweet / bitter / umami:** GPCR mechanisms.
- **Glomeruli encode odour patterns.** They are in the olfactory bulb, not the thalamus.



Audition and vestibular sensation belong to the special senses; the detailed mechanisms developed here are somatic sensation, vision, taste and smell.